User Workflow

Uncertainty-Aware Emergency Route and ETA Prediction System

1. Overview

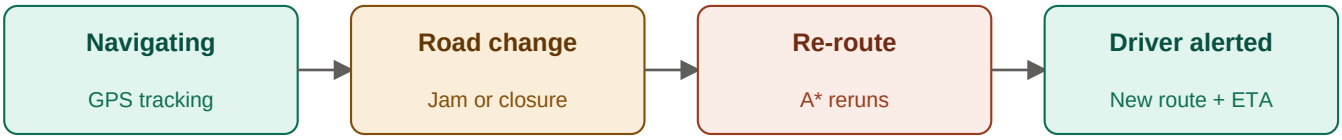
This document describes how a user interacts with the system from the moment an emergency trip begins until arrival at the hospital. It covers the main journey, the automatic rerouting loop that runs during navigation, and a separate non-urgent mode for planning the best departure time. Each step is mapped to the component of the technology stack that handles it.

2. Main user journey

Step	User action	System behaviour	Components
1. Set location	Shares GPS position or types a start point.	Snaps the location to the nearest node on the road graph.	React.js, FastAPI, OSMnx
2. Pick hospital	Accepts the suggested nearest hospital or selects another.	Loads hospital locations and candidate destinations.	OSM data, PostgreSQL
3. Route options	Compares the fastest and the most reliable route.	Predicts travel time per road segment, then runs A* for each objective.	XGBoost / LightGBM, A*
4. Review ETA	Reads the arrival range, e.g. 12-18 min, for each route.	Returns lower and upper bounds (e.g. 10th and 90th percentile) from quantile regression.	Quantile regression
5. Navigate	Follows the selected route on the live map.	Tracks position and displays the route and remaining ETA.	Folium / Leaflet, React.js
6. Arrive	Reaches the hospital.	Stores predicted ETA, actual time and route for later evaluation (MAE, RMSE).	PostgreSQL, Scikit-learn

3. Live rerouting loop

Step 5 is not a straight line. While the user navigates, the backend keeps checking for traffic, weather and road-closure changes. If a change affects the current route, the loop below runs automatically until arrival.



Repeats until arrival

- **Road change:** a traffic jam, accident, weather event or closure is detected on the active route.
- **Re-route:** travel times for affected segments are re-predicted and A* is rerun from the current position.
- **Driver alerted:** the new route and updated ETA range are shown clearly, with the reason for the change.

4. Non-urgent mode: best departure time

For trips that are not emergencies, such as scheduled hospital appointments, the user enters a destination and a time window. The system predicts ETA ranges across that window and recommends the departure time with the lowest expected travel time. This satisfies the "suggest the best departure time" objective.

Step	Description
1	User enters destination and an earliest and latest departure time.
2	System predicts ETA ranges for departures across the window.
3	System recommends the best departure time and shows the trade-off against other times.
4	User confirms and the trip starts using the main journey flow.

5. Alternate and edge cases

Situation	Expected behaviour
GPS unavailable	Prompt for a manual start point on the map.
Selected hospital unreachable	Suggest the next-nearest reachable hospital and explain why.
Weather or traffic data missing	Fall back to historical averages and flag the ETA as lower confidence.
Very wide ETA range	Highlight the uncertainty and offer the more reliable alternative route.
User leaves the planned route	Recalculate from the current position and update the ETA.

6. Design decisions to settle

- **Definition of "reliable":** narrowest ETA range, or lowest worst-case (90th percentile) time. The worst-case definition is recommended for emergencies.
- **Update frequency:** how often changes are checked during navigation, for example every 30 to 60 seconds.
- **Alert threshold:** how large a change in ETA justifies interrupting the driver with a new route.